ADVANCED MODERN PHYSICS — COURSE SYLLABUS

Course Code: PHYS-301 | Total Marks: 100

Unit 1: Special Relativity (Weight: 25 marks)

- Michelson-Morley experiment and constancy of the speed of light.
- Lorentz transformations, time dilation, and length contraction.
- Relativistic momentum and mass-energy equivalence E=mc^2.

Unit 2: Quantum Foundations and Wave Mechanics (Weight: 35 marks)

- Photoelectric effect, work function, and Compton scattering.
- De Broglie matter waves and wave-particle duality.
- Heisenberg uncertainty principle: position-momentum and energy-time.
- 1D infinite square well: eigenfunctions and quantized energy levels.

Unit 3: Atomic Structure and Spectroscopy (Weight: 25 marks)

- Bohr model of hydrogen atom and Rydberg formula.
- Pauli exclusion principle and Hund rules.

Unit 4: Nuclear and Particle Physics (Weight: 15 marks)

- Nuclear binding energy, mass defect, and radioactive decay law.